

Complex Analysis: M. Math I: Mid Semester Examination

February 27, 2025.

Maximum Marks 40

Maximum Time 2:00 hrs.

State any result you use or assume. Brief and clearly expressed answers are appreciated. Unnecessary arguments gives negative impression.

Notation: Ω is a connected open subset of $\mathbb{C}$. $f \in H(\Omega)$ means f is holomorphic on Ω . ∂S is boundary of $S \subset \mathbb{C}$. $\mathbb{D}$ is unit disc. $\Re z, \Im z$ are real imaginary parts of z . $\sqcup$ is disjoint union.

1. A function $f \in H(\Omega \setminus \{z_2\})$ has exactly one simple zero at $z_1 \in \Omega$ and a simple pole at $z_2 \in \Omega$. Show that there exists a $g \in H(\Omega)$ which does not vanish anywhere in Ω such that

$$f(z) = \frac{z - z_1}{z - z_2} g(z)$$

for all $z \in \Omega \setminus \{z_1, z_2\}$.

[8]

2. Let f be an entire function on $\mathbb{C}$. Suppose that for some $R > 0$, $|f(z)| \leq C|z|^{3/2}$ whenever $|z| > R$. Show that $f(z) = az + b$ for some constants a, b .

[8]

3. Find number of roots (counting multiplicities) of the polynomial $f(z) = z^7 + 5z^3 - z - 2$ inside $\mathbb{D}$. State any theorem you use and verify that all hypotheses are satisfied.

[8]

4. Take $f, g \in H(\Omega)$. Let D be a disc with $\bar{D} \subset \Omega$. Suppose that f has exactly one zero at $a \in D$ of order n and f does not vanish on ∂D .

Find

$$\frac{1}{2\pi i} \int_{\partial D} \frac{f'(z)}{f(z)} g(z) dz,$$

in terms of g, a, n .

[10]

P.T.O.

$$f(z) = (z-a)^n h(z)$$

$$\frac{f'(z)}{f(z)} = \frac{n}{z-a} + \frac{h'(z)}{h(z)}$$

$$\frac{1}{2\pi i} \int_{\partial D} \frac{f'(z)}{f(z)} g(z) dz = \frac{n}{2\pi i} \int_{\partial D} \frac{g(z)}{z-a} dz + \frac{1}{2\pi i} \int_{\partial D} \frac{h'(z)}{h(z)} g(z) dz$$

5. Let $\Omega \subset \mathbb{C}$ be symmetric with respect to the line $L = \{z \in \mathbb{C} \mid \Re z = \Im z\}$ and $I = \Omega \cap L$. Let Ω^+ (resp. Ω^-) be the open subset of Ω above (resp. below) I . Thus $\Omega = \Omega^+ \sqcup I \sqcup \Omega^-$. Suppose $f \in H(\Omega^+)$ which extends as a continuous function on I . Find a sufficient condition such that f can be extended as a holomorphic function on Ω . Justify your answer. (You may assume the reflection principle done in class.) [8]

6. Take $\Omega \supset \mathbb{D}$ and a function $F \in H(\Omega)$. Let $F|_{\partial\mathbb{D}} = f$. Thus f is a function on the unit circle. Its Fourier transform $\hat{f}$ is defined as

$$\hat{f}(n) = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta}) e^{-in\theta} d\theta \quad \forall n \in \mathbb{Z}.$$

If $\hat{f}(n) = \hat{f}(-n)$ for all $n \in \mathbb{Z}$, then show that f is a constant. [8]

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